

Emotion based Story Telling App

UCS 503 Software Engineering Project Report END-Semester Evaluation



THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

Submitted By:

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BE Third Year, 3CO28

Submitted To:

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TIET, Patiala

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Project Selection Phase

1.1 Software Bid:

Team Name:

Team ID (will be assigned by Instructor):

Please enter the names of your Preferred Team Members.

- You are required to form a **three to four person** teams
- Choose your team members wisely. You will not be allowed to change teams.

Name	Roll No	Project Experience	Programming Language used
Siddharth Maithani	102153035	Bank Management System, AI based Mancala Game, Hand Gesture Detection.	Python, Java, C++, Sql, javascript
Kartik Kumaria	102103830	Bank Management System, AI based Mancala Game, Hand Gesture Detection.	Python, C++, Sql, javascript, PHP
Aalok Kumar Yadav	102103846	Hostel Management System, Real Time Color Detection.	Python, Sql

Programming Language / Environment Experience

List the languages you are most comfortable developing in, **as a team**, in your order of preference. Many of the projects involve Java or C/C++ programming.

1. Python
2. Javascript
3. HTML/CSS
4. ReactJs

Choices of Projects:

Please select **4 projects** your team would like to work on, by order of preference:

[Write at-least one paragraph for each choice (motivation, reason for choice, feasibility analysis, etc.)]

First Choice	Emotion Based Story Telling App
Second Choice	Emotion Based Music Recommender
Third Choice	Handwritten Notes to Text converter
Fourth Choice	Language Translator for websites

1.2 Project Overview

Emotion Based Story Telling App

Objective

The objective of our emotion-based storytelling app is to deliver personalized and engaging narratives that dynamically adapt to the user's emotional state. By leveraging advanced emotion detection technology, the app aims to intuitively understand and respond to the user's feelings in real-time. Our goal is to create a unique storytelling experience, tailoring content to evoke emotions such as joy, excitement, or empathy. Through a seamless integration of emotional intelligence and captivating narratives, we aspire to offer users a novel and immersive storytelling journey that transcends traditional boundaries, fostering a deeper connection between the audience and the content.

Scope of Application

The objective of our emotion-based storytelling app is to deliver personalized and engaging narratives that dynamically adapt to the user's emotional state. By leveraging advanced emotion detection technology, the app aims to intuitively understand and respond to the user's feelings in real-time. Our goal is to create a unique storytelling experience, tailoring content to evoke emotions such as joy, excitement, or empathy. Through a seamless integration of emotional intelligence and captivating narratives, we aspire to offer users a novel and immersive storytelling journey that transcends traditional boundaries, fostering a deeper connection between the audience and the content.

Product Features

- **Real-time Emotion Detection:**

The app employs advanced emotion recognition technology to analyze user facial expressions and adjust storytelling content dynamically based on detected emotions.

- **Personalized Narratives:**

Tailored storytelling experiences adapt to the user's emotional state, providing a unique and personalized journey with narratives designed to evoke specific feelings such as joy, excitement, or empathy.

- **Interactive User Engagement:**

Users can actively influence the storyline by expressing emotions, fostering an interactive and immersive experience that goes beyond conventional passive consumption of content.

- **Multi-Genre Storytelling:**

The app offers a diverse range of stories spanning various genres, ensuring a wide appeal and catering to different emotional preferences and moods of the users

- **Privacy and Security Measures:**

The project prioritizes user privacy by implementing robust security measures for handling sensitive emotional data, adhering to ethical standards, and providing transparency in data usage practices.

Time Of Completion

Before End Semester Test (Around Academic Week 15/16) Odd Semester,
Academic Year – 2023

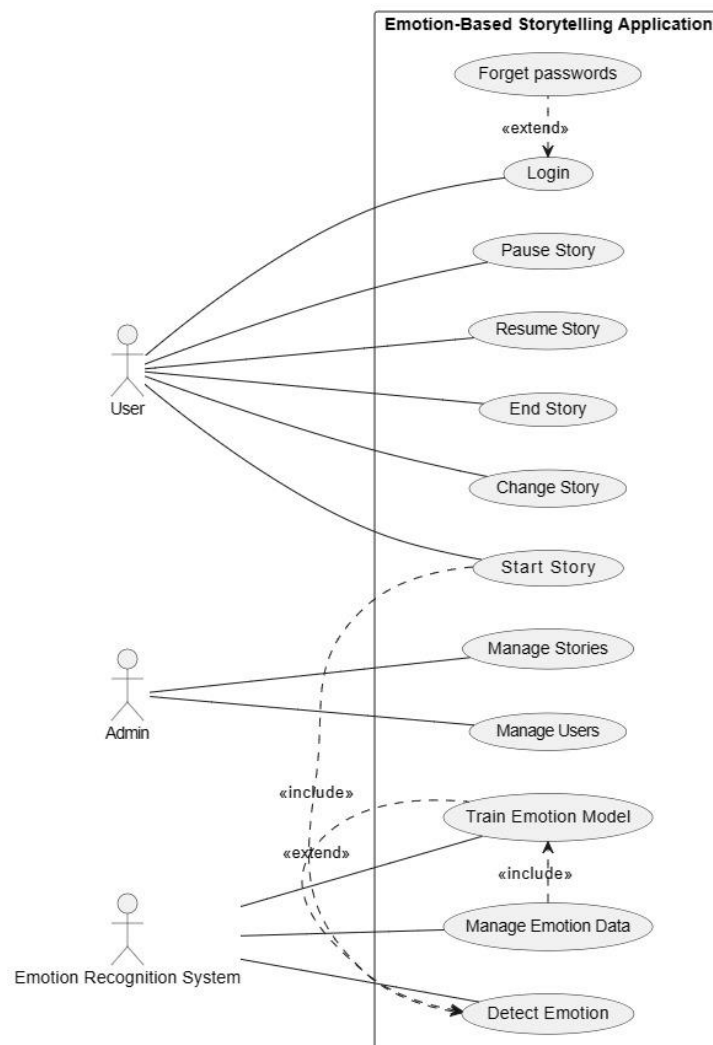
Challenges

- Building an interactive and responsive UI with the latest design practices.
- Making an Android Application for faster user experience.
- Developing a machine learning model to detect user emotion.
- Generating Stories that corresponds to user real time emotions.

2. Analysis Phase

2.1 Use Cases

2.1.1 Use Case Diagram



2.1.1 Use Case Template

2.1.1.1 Template for creating an account

1. Use Case Title	Create Account
2. Abbreviated title	Create Account
3. Use Case ID	101
4. Actors	User
5. Description - User: Wants to create an account to access the website's features and services. - Website Owner: Wants to provide a user-friendly and efficient account creation process to attract and retain users.	
5.1. Preconditions: - User has access to the internet and a web browser. - User has not previously created an account on the website.	
5.2. Task Sequences/Normal scenario: - User Navigates to SignUp Page - User enters their Name,Email,Mobile Number,Password to setup their account - The User Clicks SignIn - The System validates the user details - The system redirects the user to Log In Page	
5.3. Alternative scenario: If the user enters invalid or incomplete information, the system displays an error message and prompts the user to enter valid information.	
5.4. Post condition: The User can continue with login process	
6. Modification History: Date 25 Nov 2023	
7. Author: Kartik Kumaria	

2.1.1.2 Template for Login in to user account

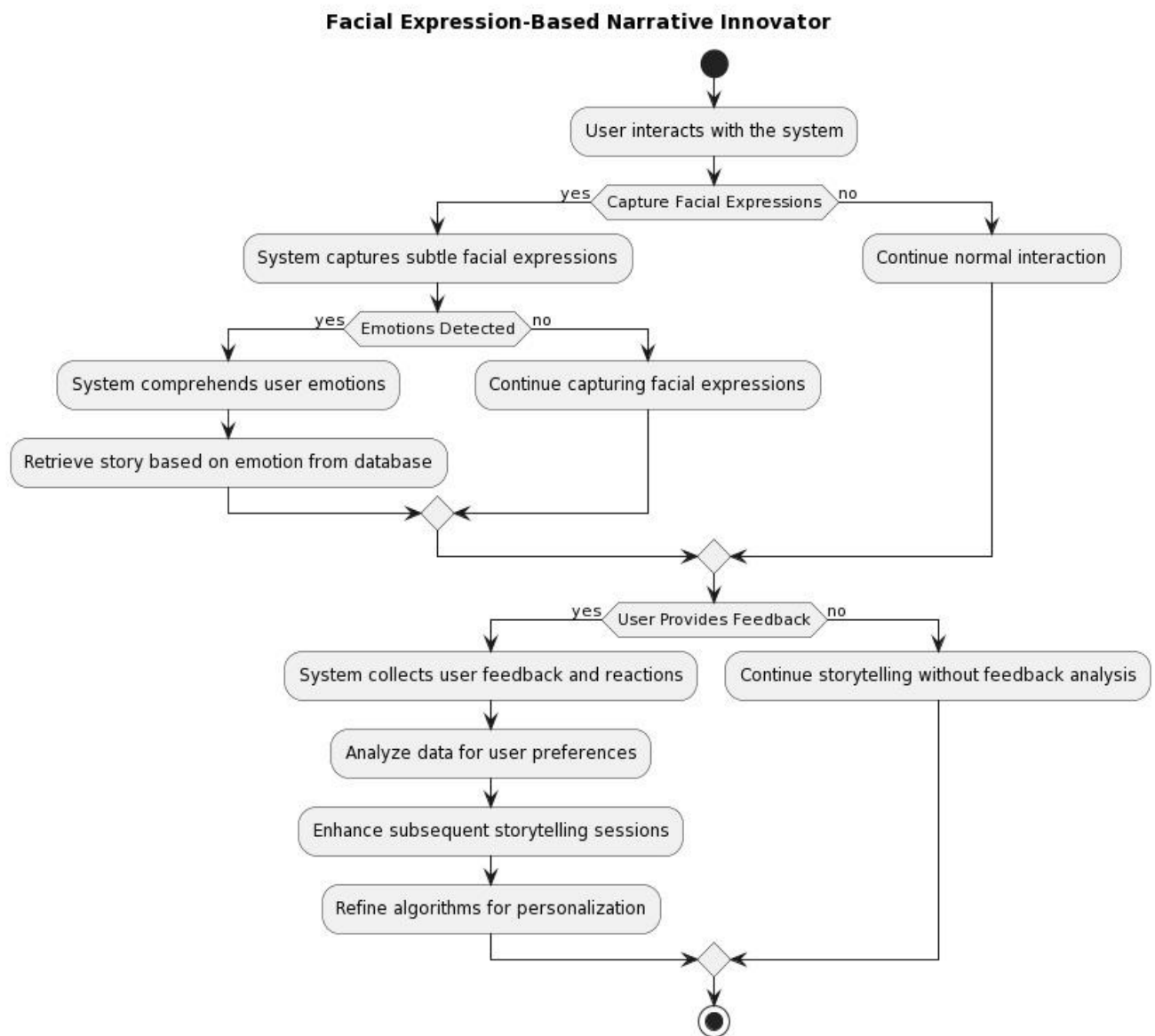
1. Use Case Title	Log In
2. Abbreviated title	Log In
3. Use Case ID	102
4. Actors	User
5. Description	- User: Wants to create an account to access App's Features. - Website Owner: Wants to provide a user-friendly and efficient Log In process to attract and retain users.
5.1. Preconditions:	- User has access to the internet and a web browser. - User has already created their account
5.2. Task Sequences/Normal scenario:	- User Navigates to Login Page - User enters their Email/Mobile as User ID and their Password - The User Clicks LogIn - The System validates the user credentials - The system redirects the user to Entry Page
5.3. Alternative scenario:	If the user enters invalid or incomplete information, the system displays an error message and prompts the user to enter valid information.
5.4. Post condition:	The User can access website features
6. Modification History: Date 25 Nov 2023	
7. Author: Siddharth Maithani	

2.1.1.3 Template for Displaying Story

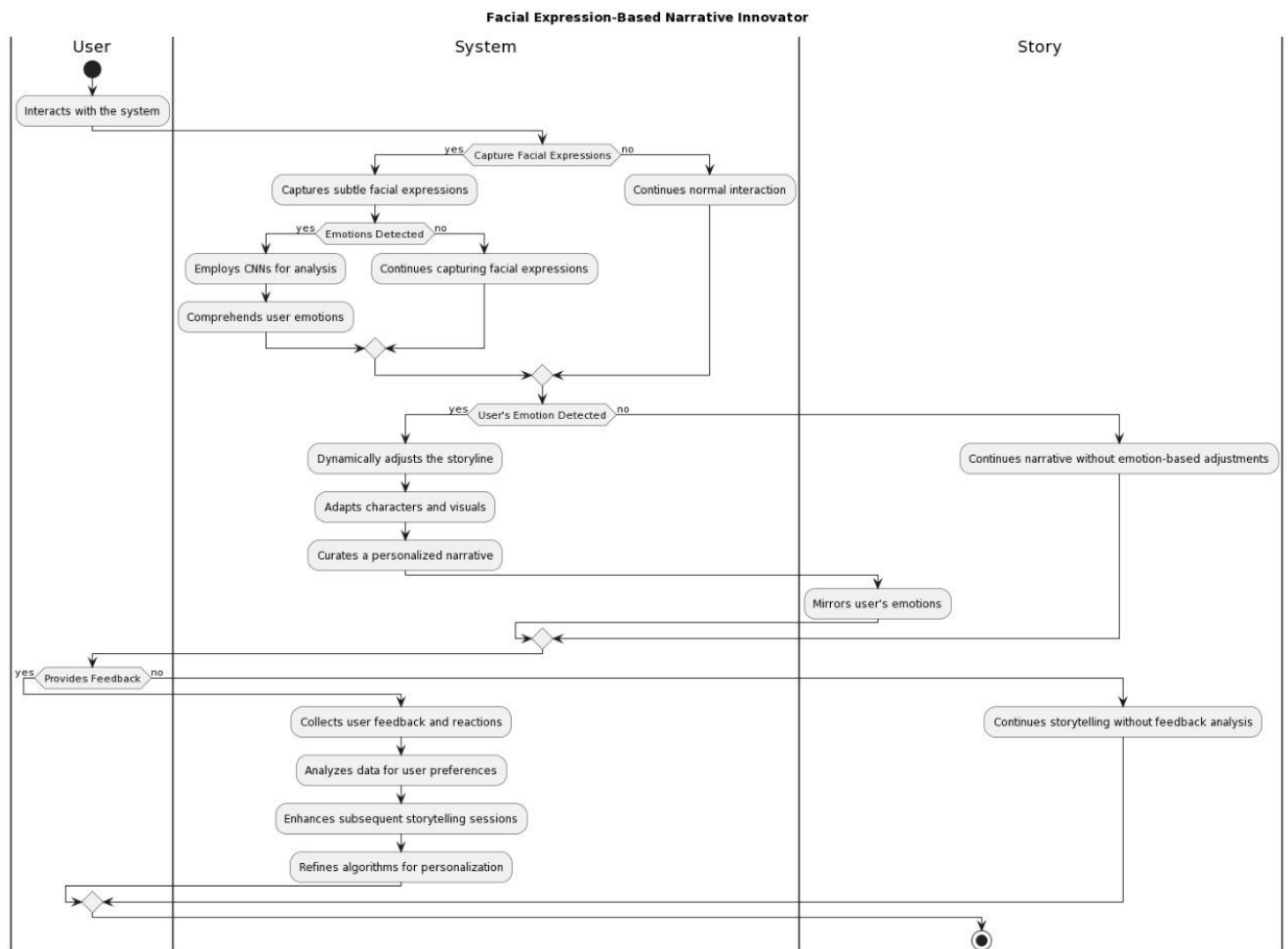
1. Use Case Title	Story Display
2. Abbreviated title	Story Display
3. Use Case ID	103
4. Actors	User
5. Description	- User: Wants to read stories based on their mood. - Website Owner: Wants to provide Users with stories for their mood to improve their experience.
5.1. Preconditions:	- User has an active account on the website. - The emotion of User has been predicted
5.2. Task Sequences/Normal scenario:	- User is Shown their detected emotion - User clicks on Show Stories - A random Story based on the emotion of the user is generated from the database - The system passes the story to the StoryTeller which displays the story to user
5.3. Alternative scenario:	If no emotion is detected/Faceless picture is detected, then, The User is alerted about it, and taken back to the face capture process
5.4. Post condition:	User Can enjoy the generated story with an option to regenerate a new story according to their liking
6. Modification History: Date 25 Nov 2023	
7. Author: Aalok Kumar Yadav	

2.2 Activity/Swimlane Diagram

2.2.1 Activity Diagram

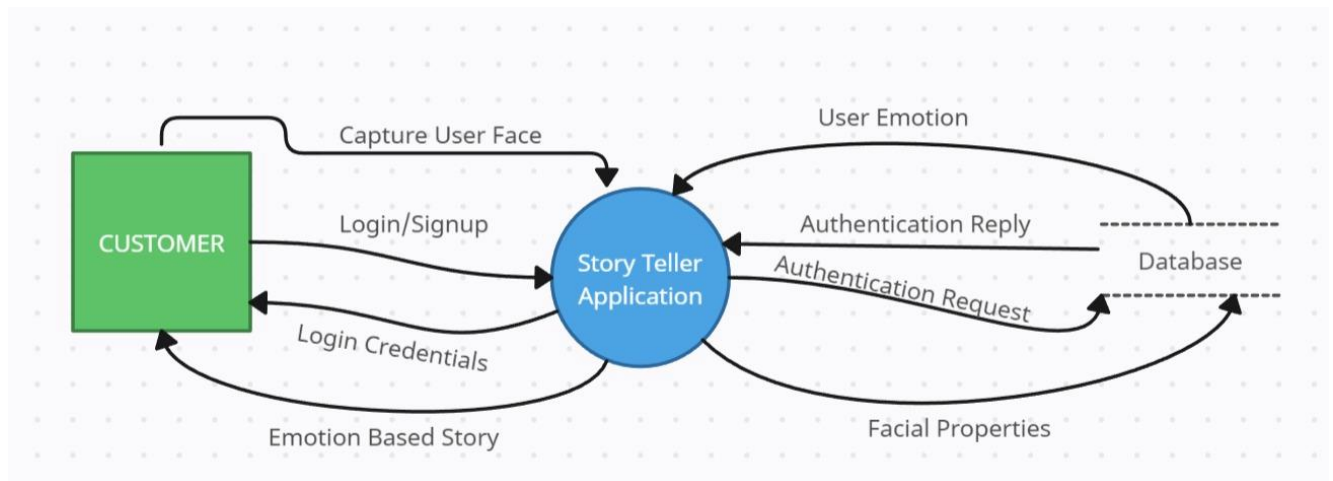


2.2.2 Swimlane Diagram

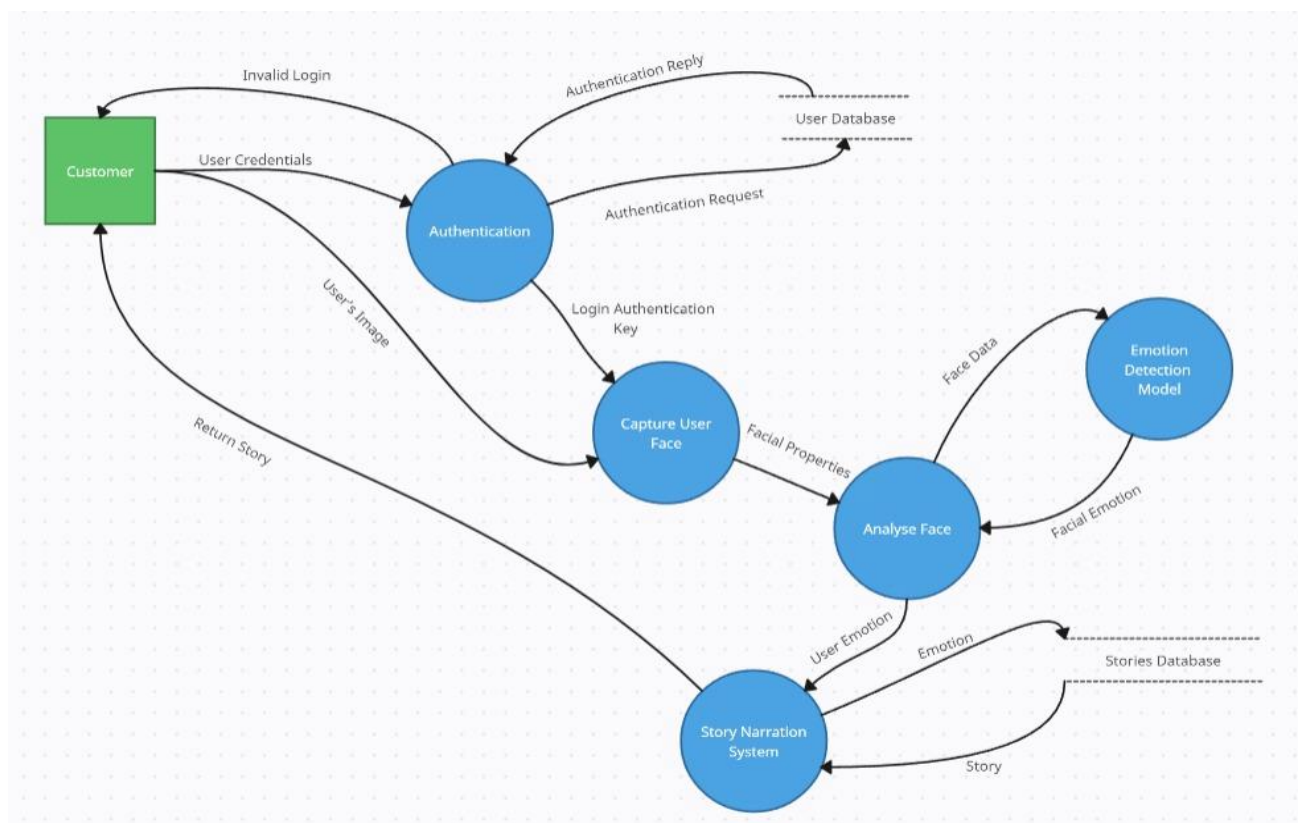


2.3 Data Flow Diagrams(DFD's)

2.3.1 Level 0 Data Flow Diagram



2.3.2 Level 1 Data Flow Diagram



2.3 Software Requirement Specification in IEEE Format

Software Requirements Specification Document

Version 1.0

Emotion Based Story Telling App

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1 Introduction

1.1 Purpose of this Document

The purpose of this SRS document is to provide a comprehensive understanding of the facial emotion based storytelling webapp. It outlines the key parameters, goals, and requirements of the software product. The document also includes details on the target audience, user interface, and hardware and software specifications. It serves as a guideline to ensure that the client's expectations, team's vision, and user's needs align with the product's functionality. Overall, this SRS document aims to provide a clear and detailed overview of the project to all stakeholders involved.

1.2 Scope of the Project

- The project's objective is to craft a storytelling application that harnesses the capabilities of facial emotion recognition technology. The primary aim is to design an engaging and interactive storytelling platform that adapts to the user's emotional responses in real-time.
- Utilizing sophisticated facial emotion recognition algorithms, the application will identify and interpret the user's facial expressions. This data will be utilized to tailor the storyline, modifying the narrative's direction, characters, and dialogues based on the user's emotional expressions. For instance, if the user displays sadness or fear, the story may shift towards a more uplifting or comforting narrative.
- The project entails various crucial phases, including the development of the facial emotion recognition system, the creation of a dynamic storytelling database, and the seamless integration of these components into a user-friendly application. Extensive testing will be conducted to ensure precise emotion detection and a smooth storytelling experience.
- This application has the potential to redefine digital storytelling, offering a more personalized and captivating encounter. Its applications extend to diverse fields like entertainment, education, therapy, and beyond. The project signifies an exciting convergence of technology and creativity, leveraging artificial intelligence to construct narratives that are truly responsive and immersive.

1.3. Definitions, Abbreviations, and Acronyms

Table 1 gives the explanation of the most commonly used terms in this SRS document

Table 1: Definitions for most commonly used terms

S.No.	Term	Definition
1.	Facial Emotions	Facial expressions are part of our instinctual body language that reveal our emotional states outwardly. They are often subtle and can change our gaze, smile, or use other microexpressions specific to an emotion.
2.	Data Structure & Algorithms	A data structure is a method of organizing data in a virtual system. Think of sequences of numbers, or tables of data: these are both well-defined data structures. An algorithm is a sequence of steps executed by a computer that takes an input and transforms it into a target output.
3.	Node.js	As an asynchronous event-driven JavaScript runtime, Node.js is designed to build scalable network applications.
4.	Express	Express is a minimal and flexible Node.js web application framework that provides a robust set of features for web and mobile applications.
5.	React Js	React is a free and open-source front-end JavaScript library for building user interfaces based on components.
9.	Computer Vision	Computer vision is a field of study that focuses on enabling machines to interpret and understand visual data from the world around them.

Abbreviations

Table 2 gives the full form of most commonly used mnemonics in the SRS document

Table 2: Full form for most commonly used mnemonics

S.No.	Mnemonic	Full Form
1.	DSA	Data Structure and Algorithms
2.	HTML	Hypertext Markup Language
3.	CSS	Cascading Style Sheets
4.	JS	JavaScript

1.4. References

OpenCV documentation- <https://opencv.org/>

Node.js documentation - <https://nodejs.org/en/docs>

HTML documentation - <https://developer.mozilla.org/en-US/docs/Web/HTML>

CSS documentation - <https://developer.mozilla.org/en-US/docs/Web/CSS>

JavaScript documentation - <https://developer.mozilla.org/enUS/docs/Web/JavaScript>

React JS documentation - <https://react.dev/learn>

Express documentation - <https://expressjs.com/>

1.5 Overview

The remaining sections of this document provide a general description, including characteristics of the users of this project, the product's hardware, and the functional and data requirements of the product. General description of the project is discussed in section 2 of this document. Section 2 gives the functional requirements, data requirements and constraints and assumptions made while designing the multi-utility system. also gives the user viewpoint of product use. Section 3 gives the specific requirements of the product. Section 3.0 also discusses the external interface requirements and gives detailed description of functional requirements.

2. Overall Description

2.1. Product Perspective

The innovative storytelling app employing facial emotion recognition technology is a groundbreaking advancement set to transform the realm of digital storytelling. Through the integration of advanced emotion recognition tech, this app delivers a distinctive, tailor-made storytelling venture that dynamically responds to the user's emotional cues in real-time.

At its core, the app excels in interpreting the user's facial expressions by employing intricate emotion recognition algorithms. This information is then utilized to adjust the story's plot, character interactions, and dialogues, aligning seamlessly with the user's emotional reactions. This dynamic storytelling approach ensures an enthralling and immersive experience that deeply connects with the user.

From a product standpoint, this app epitomizes technological innovation and imaginative storytelling, melding artificial intelligence with narrative creativity to offer an authentically responsive storytelling journey. Its potential applications span across various domains, including entertainment, education, therapy, and more.

Furthermore, the app's user-friendly interface guarantees a smooth and accessible user experience, making it widely approachable. Rigorous testing will be conducted to ensure precise emotion detection and fluid narrative transitions.

In summary, this storytelling application marks a significant leap in the domain of digital storytelling. By providing interactive narratives that adapt to the user's emotions, it presents a profoundly personal and captivating storytelling experience that distinguishes it in the market.

2.2. Product Functions

The project titled Facial Emotion based Story Telling app strives to craft an engaging storytelling venture through the utilization of facial emotion detection technology. Employing convolutional neural networks (CNNs), the system effectively identifies and interprets real-time facial expressions¹. By scrutinizing the user's facial attributes, the system can accurately recognize emotions such as joy, sorrow, ire, apprehension, and astonishment¹.

Key functionalities of the Facial Expression-Based Narrative Innovator encompass:

1. **Real-time Emotional Discernment:** The system captures subtle facial expressions and employs CNNs for immediate analysis². This empowers the system to precisely detect and comprehend emotions, culminating in a deeply engaging storytelling involvement.
2. **Interactive Storytelling:** Tailored to the user's detected emotions, the system dynamically adjusts the storyline, characters, and visuals, curating a personalized narrative journey¹. This ensures the story mirrors the user's emotional disposition, elevating engagement and immersion.
3. **Flexible Story Progression:** The Facial Expression-Based Narrative Innovator adapts its plotline based on the user's emotional reactions throughout the narrative¹. By constantly monitoring the user's facial expressions, the system can alter the story's development, pace, and mood to sustain engagement and emotional connection.
4. **Visual Augmentations:** The system incorporates dynamic visual enhancements, encompassing animations, effects, and filters, to complement the storytelling expedition. These visual elements dynamically shift based on the user's recognized emotions, further involving them in the narrative.
5. **User Feedback Analysis:** The Facial Expression-Based Narrative Innovator assesses user feedback and emotional reactions to enhance subsequent storytelling sessions. By accumulating data regarding user preferences and emotional engagement, the system refines its algorithms to offer more personalized and emotionally resonant narratives.

The Facial Expression-Based Narrative Innovator amalgamates advanced facial emotion recognition technology with interactive storytelling, fashioning a distinctive and captivating user experience. By harnessing real-time emotion detection and adaptive storytelling methodologies, this initiative aspires to reshape conventional storytelling approaches, delivering immersive tales tailored to each user's emotional state.

2.3 User Characteristics

The goal is to design a software that would detect the real time emotions of a user and then generate and recommend certain short stories according to his emotion detected.

Thus, while designing the software one can assume that each user type has the following characteristics

- The user should be open to personalized content based on their emotions. They should be comfortable with the app tailoring stories that cater to their emotional state
- The user should be comfortable using technology, specifically the app's interface. They should be able to navigate the app easily, input their emotional feedback, and understand the app's features
- The user should be able to express their emotions accurately to the app
- As the app generates stories based on user emotions, the user should be patient enough to complete the stories and curious to explore different storylines that the app generates

2.4 Apportioning of requirements

The Emotion based story telling app is to be implemented in the following phases:

- **Beta Phase:** The first prototype of the software will be deployed on a private system that would only be accessible to limited user and developers only.
- **Android/IOS Deployment:** After the prototype is tested with needed verification and updates, the software will be open for users from world-wide to have an handy application that would generate custom stories for them.

3. Specific Requirements

3.1 External Interface Requirements

The following list presents the external interface requirements

- The product requires very limited graphic usage with just a simple interface for capturing your photo through the device camera as user input.

- The product has an responsive UI, meaning that it is compatible for most of the devices.
- The product is developed in React Native environment thus it cross platform based i.e it can be used on Android, Windows and IOS supporting devices

3.2 Detailed Description of Functional Requirements

3.2.1 Functional Requirement for user registration and login

Purpose	This function allows users to create an account and login to the website.
Input	User's Fullname,Email,MobileNumber,Password
Process	The system verifies the user's details, creates a user profile, and stores the user's information securely.
Output	User is registered and can log in to the website.

3.2.2 Functional Requirement for user Choosing a story for all stories

Purpose	This Function Allows User to Choose from All the Stories in the database
Input	User selects Explore Stories.
Process	All stories from the Database are generate and the user selects according to preference
Output	Selected Story is Displayed

3.2.3 Functional Requirement for capturing the user face

Purpose	Capture the User Face
Input	User's Face
Process	The System Uses React native vision camera to capture the user image using android interface
Output	The Image is sent to model for prediction

3.2.4 Functional Requirement for predicting User Emotion

Purpose	Predict the User Emotion
Input	Image of the User's face
Process	the image is sent to the pretrained model to predict the facial emotion displayed by the user
Output	The facial emotion of the user is displayed

3.2.5 Functional Requirement for displaying story based on emotion

Purpose	Display Story Based on EMotion
Input	User's Emotion
Process	The emotion predicted is sent to the databse for story selection.the selected story is then sent to the storyTeller for display
Output	The story based on user's emotion is displayed

3.3 Performance Requirements

- **Responsiveness:** The app should respond to user inputs promptly. This includes the time it takes for the app to generate a story based on the user's emotion, as well as the time it takes for the app to update the story as the user's emotion changes
- **Usability:** The app should be easy to use and navigate. This includes the ease of understanding the content, the ease of handling the device during a task, and the ease of getting familiar with the product.
- **Reliability:** The app should work consistently and without failures. This includes the app's ability to perform its functions reliably under different conditions, as well as its ability to recover from failures
- **Security:** The app should protect user data and privacy. This includes the app's ability to securely store and transmit user data, as well as its compliance with data protection regulations

3.4 Logical Database Requirements

The logical database requirements for an emotion-based storytelling app can be outlined as follows:

- **User Profile Database:** This database would store information about each user, such as their name, email address, and preferences. It would also store the user's past emotional states and the stories they have interacted with
- **Emotion Detection Database:** This database would store the emotions detected by the app. This could include a record of each emotion detected, the time it was detected, and the user who experienced the emotion. This data could be used to improve the app's emotion detection algorithm
- **Story Generation Database:** This database would store the stories generated by the app. It would include a record of each story, the emotions it was designed to cater to, and the user who interacted with the story. This data could be used to improve the app's story generation algorithm

3.4 Quality Attributes

- **Performance:** The system should be responsive and have low latency, especially when accessing the database or third-party APIs.
- **Reliability:** The system should be reliable and have a high uptime, with backups and redundancy to ensure data integrity and availability.
- **Security:** The system should be secure and protect user data from unauthorized access, hacking attempts, and other security threats. The system should use encryption and best practices for authentication and authorization.
- **Scalability:** The system should be able to handle increasing numbers of users and data without compromising performance or reliability.
- **Compatibility:** The system should be compatible with a variety of devices, browsers, and platforms to ensure that users can access it from anywhere.
- **Testability:** The system should be designed with testing in mind, with clear unit tests and integration tests. The system should also be easy to deploy and roll back in case of errors or issues.

3.5 Other Requirements

None at this time

4 Change History

Id	Comment
200209	Version 1.0- Initial Release
200210	Version 1.1- Refined the Emotion Detection Model

5. Document Approvers

SRS for Emotion Based Story Telling App

Approved By:

Dr. Manish Kumar

Computer Science and Engineering Department

TIET, Patiala

November 2023

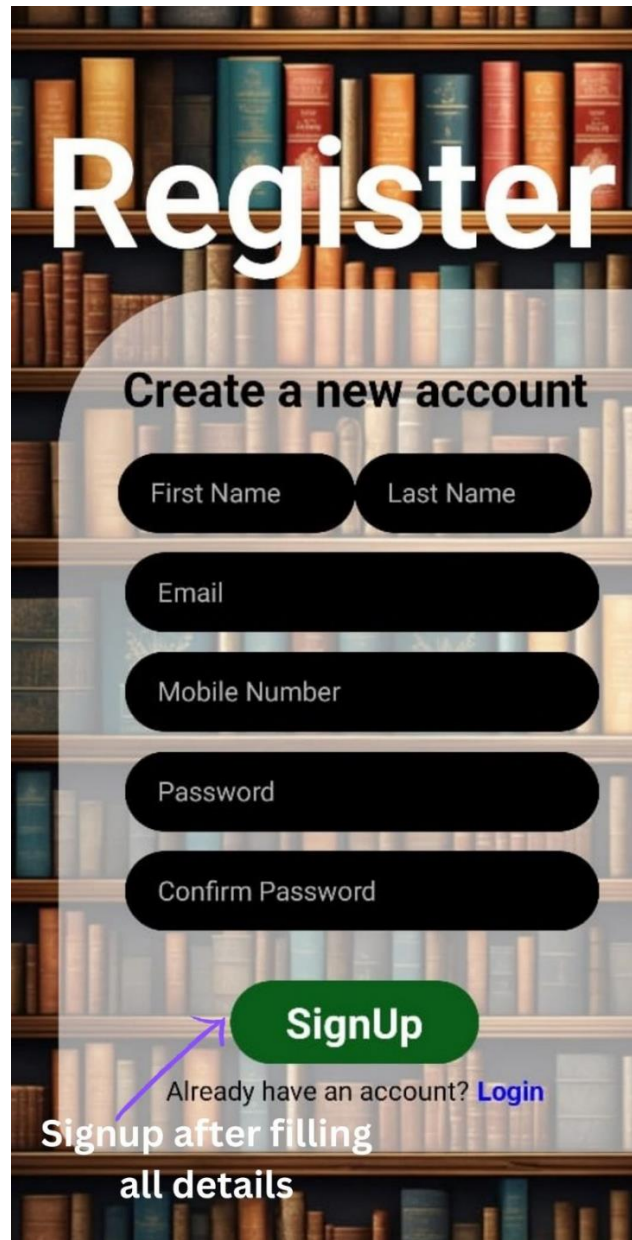
Lab Instructor

Date: 28/11/23

2.5 User Stories and Story Card

#001 User SignUp

As a user, I want to be able to create an account on the platform so that I can get personalised recommendation and get access to many more features of the app

A mobile app registration screen with a background of bookshelves. The word "Register" is at the top in large white font. Below it is the heading "Create a new account". The form contains input fields for "First Name", "Last Name", "Email", "Mobile Number", "Password", and "Confirm Password". A green "SignUp" button is at the bottom. A purple arrow points to the button with the text "Signup after filling all details". Above the button is the text "Already have an account? Login" with "Login" in blue.

Register

Create a new account

First Name Last Name

Email

Mobile Number

Password

Confirm Password

SignUp

Already have an account? [Login](#)

Signup after filling all details

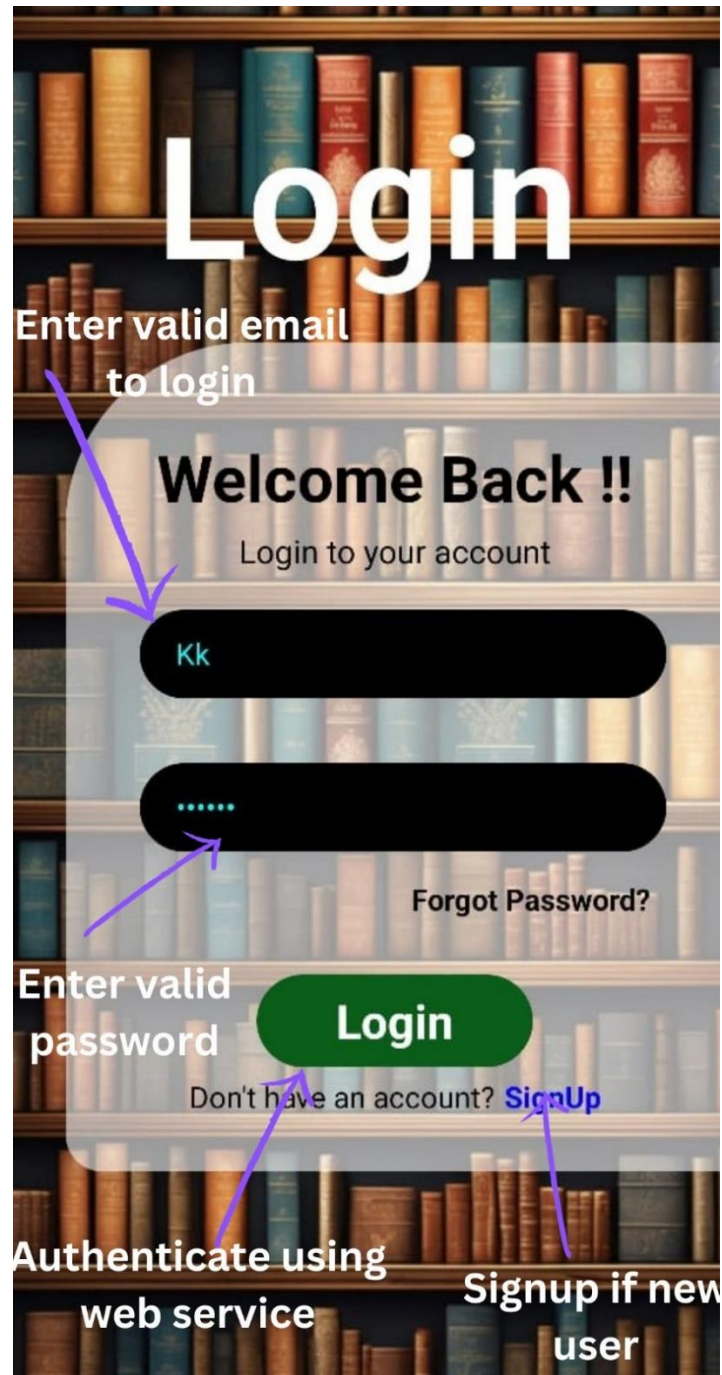
Confirmation:

Success: “Registration Successful!! Login to continue”

Failure: “Registration Failed”

#002 User Login

As a user Login page allows the user to log in to the application using his username and password



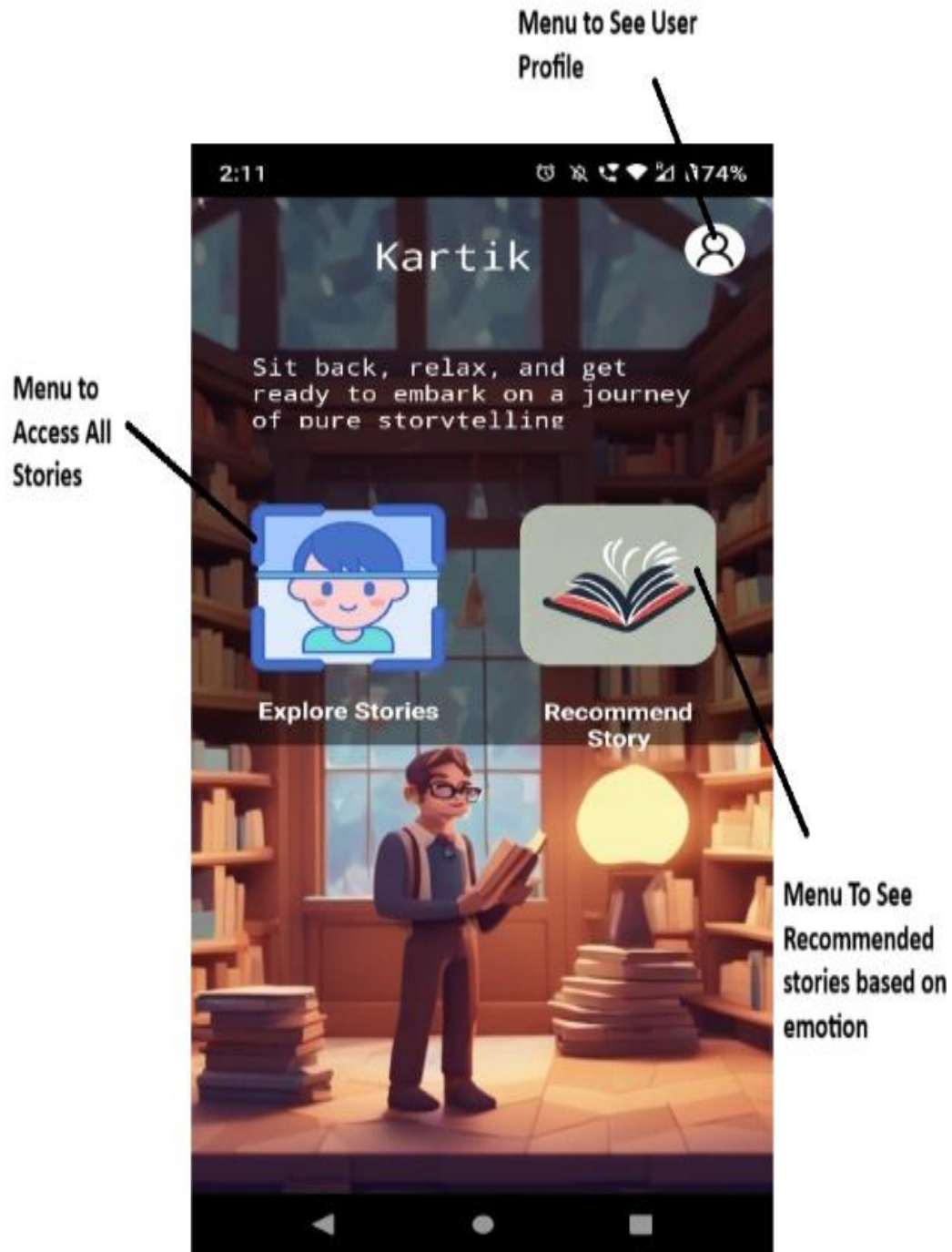
Confirmation:

Success: "Login Success"

Failure: "Invalid Username/Password"

#003 Home Page

Allows user to navigate to different sections of the project



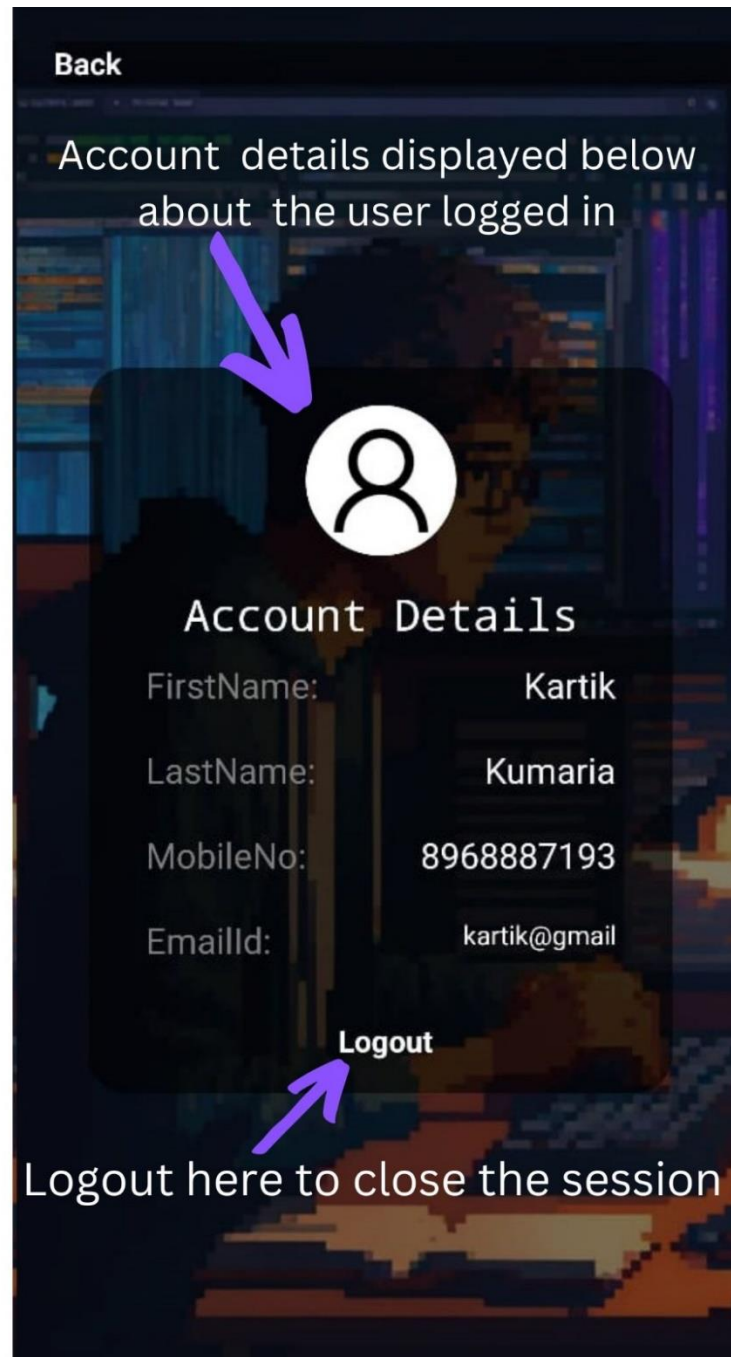
Confirmation:

Success: Navigation to required sections of the application

Failure: Unable to load the application

#004 Account Details Page

Displaying the Account Details of the user that is logged in to the application currently



Confirmation:

Success: All Details displayed successfully

Failure: Unable to fetch data from the database

#005 Emotion Capturing

Capturing User Image for processing to detect the emotion from it.

React
Native
Vision
Camera



Button to capture the image

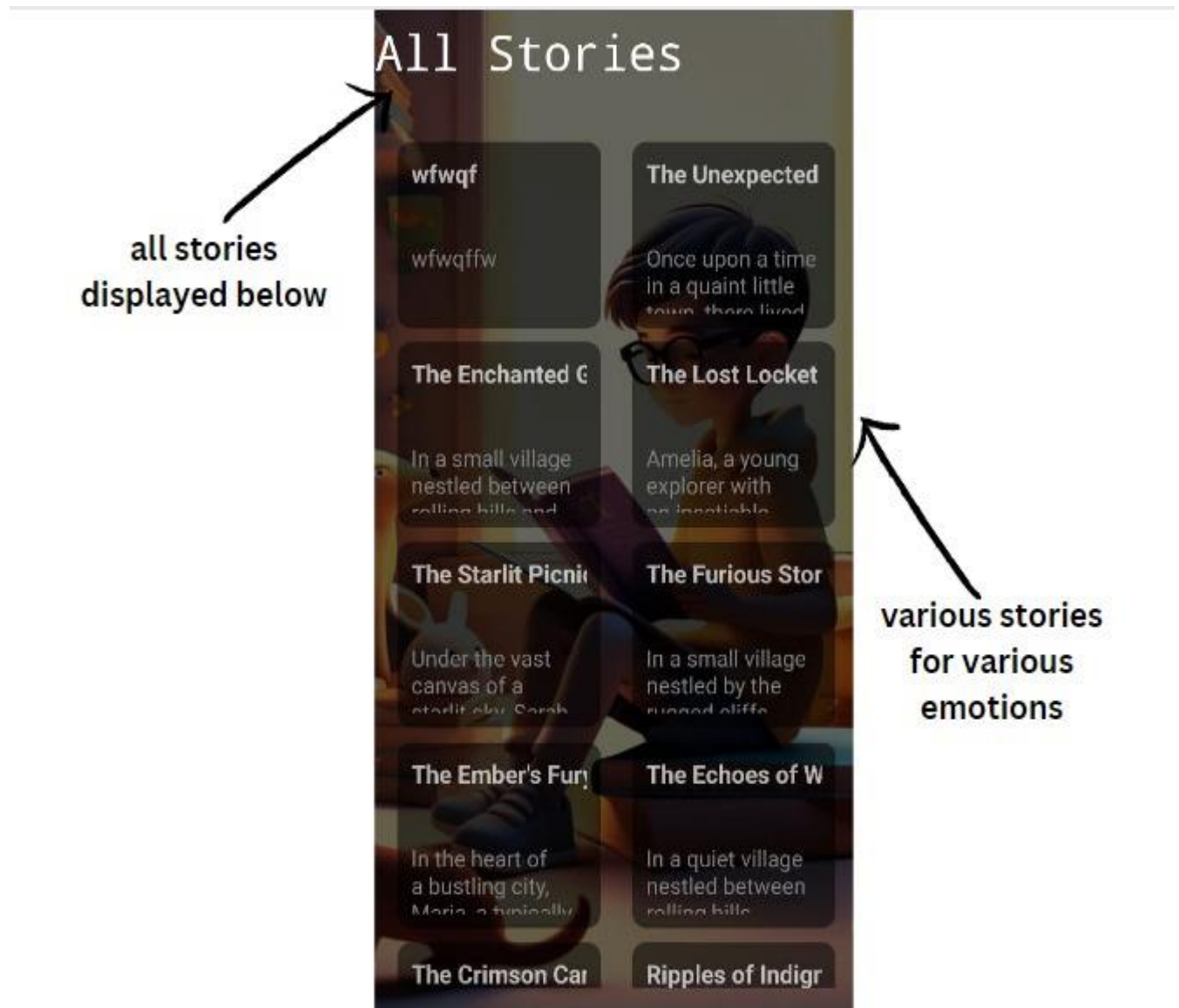
Confirmation:

Success: Image successfully captured and emotion detected

Failure: Failed to capture the image

#006 Displaying Stories

Displaying all the stories present in the stories database to



Confirmation:

Success: All stories being displayed successfully

Failure: Unable to fetch story data from the database

#007 Displaying a story

Story is displayed along with its Title to be read by the user



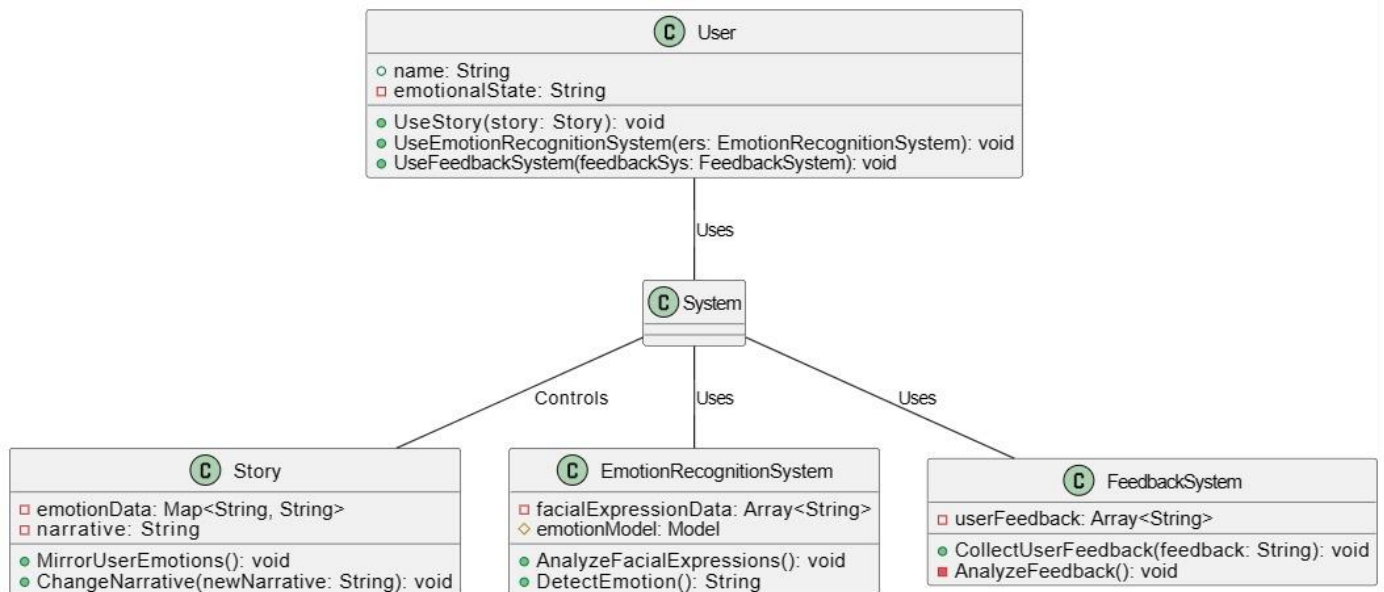
Confirmation:

Success: Entire Story is displayed on the screen to be read along with the title

Failure: Unable to fetch story from the database

3. Design Phase

3.1 Class Diagram

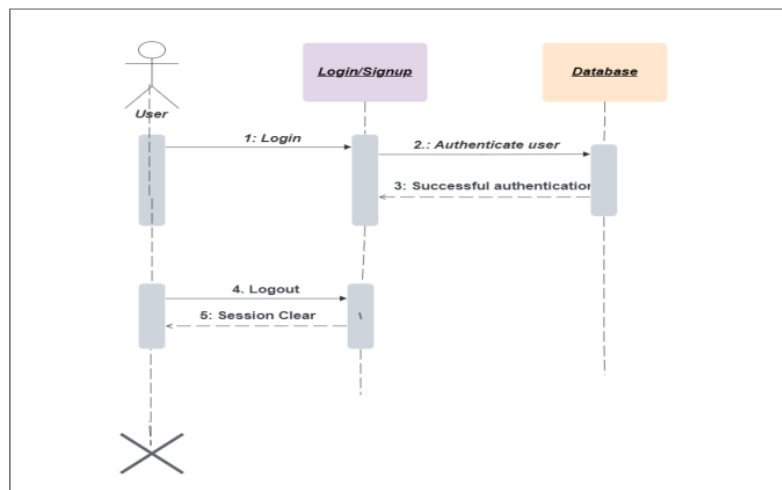


Legend:

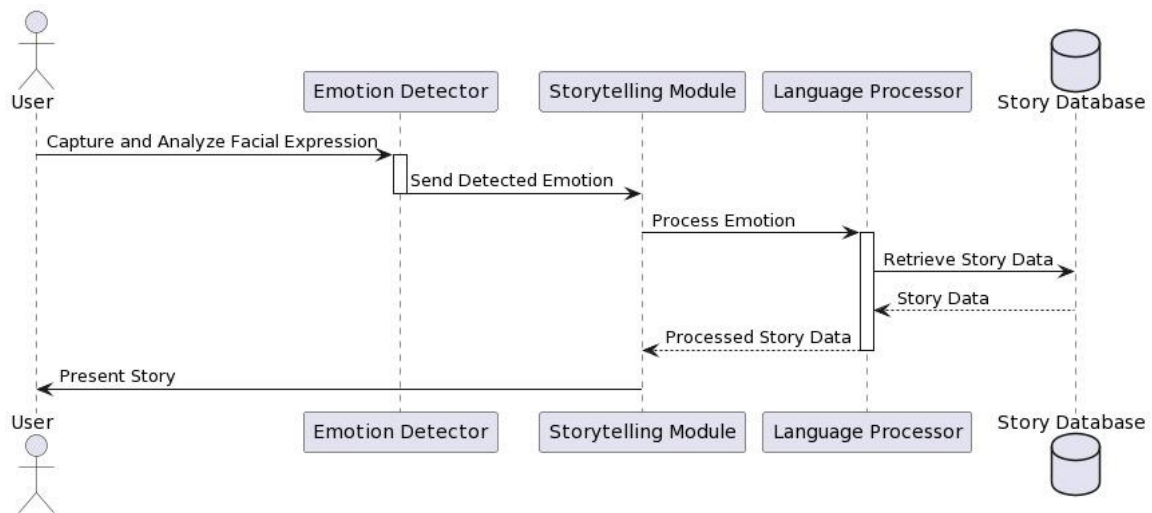
- Private
- ◇ Protected
- Public

3.2 Sequence Diagram

3.2.1 Login/SignUp

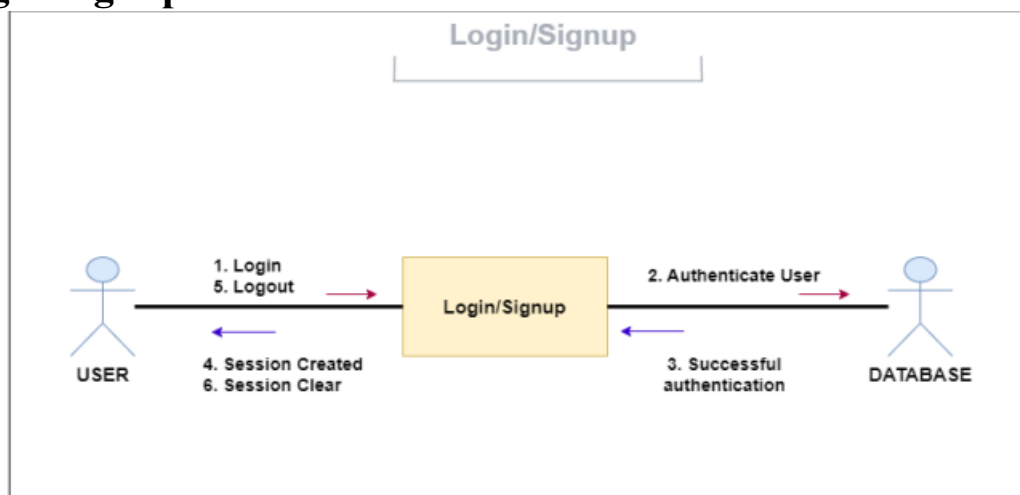


3.2.2 Emotion Detection and Story Generation

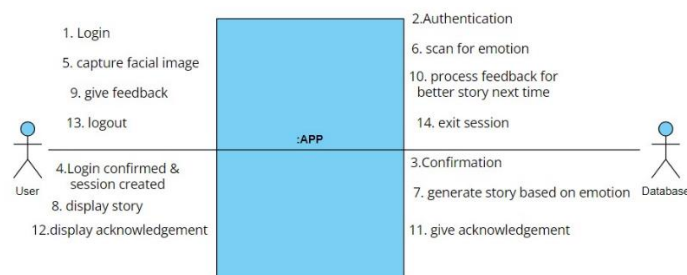


3.3 Collaboration Diagram

3.3.1 Login/Signup

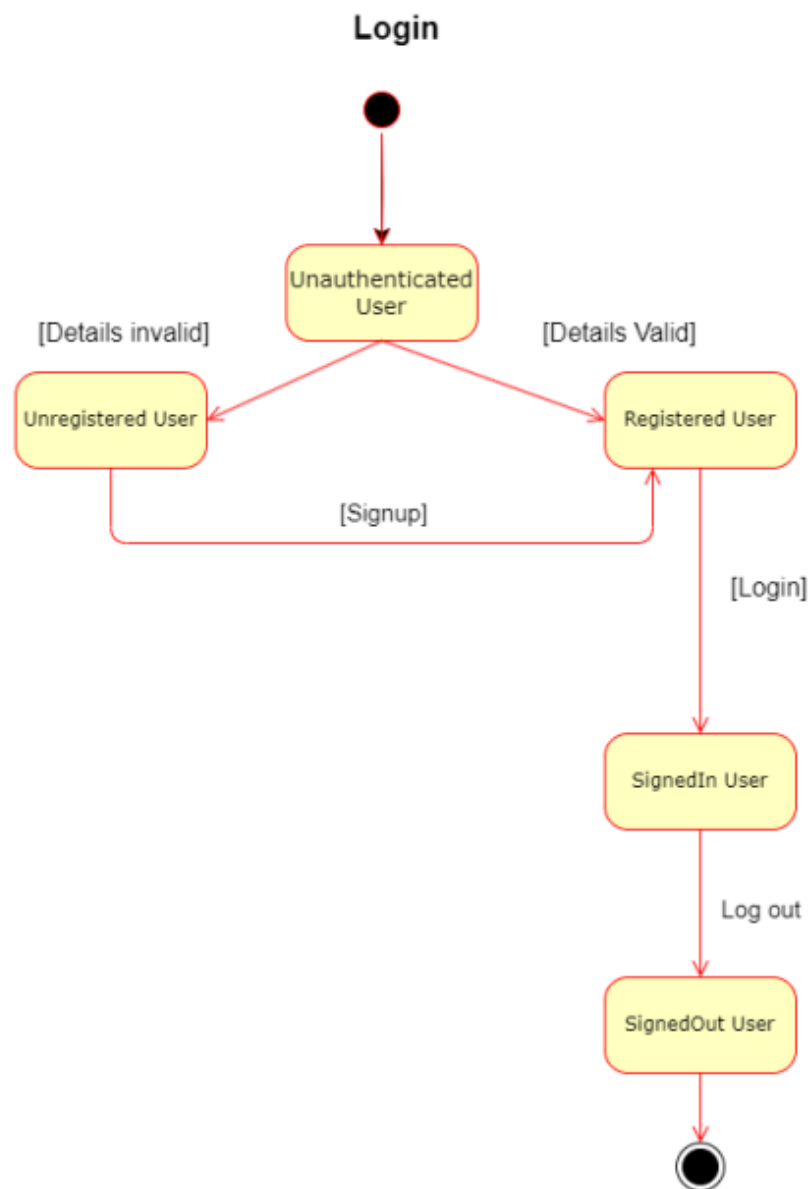


3.3.2 Emotion Detection and Story Display



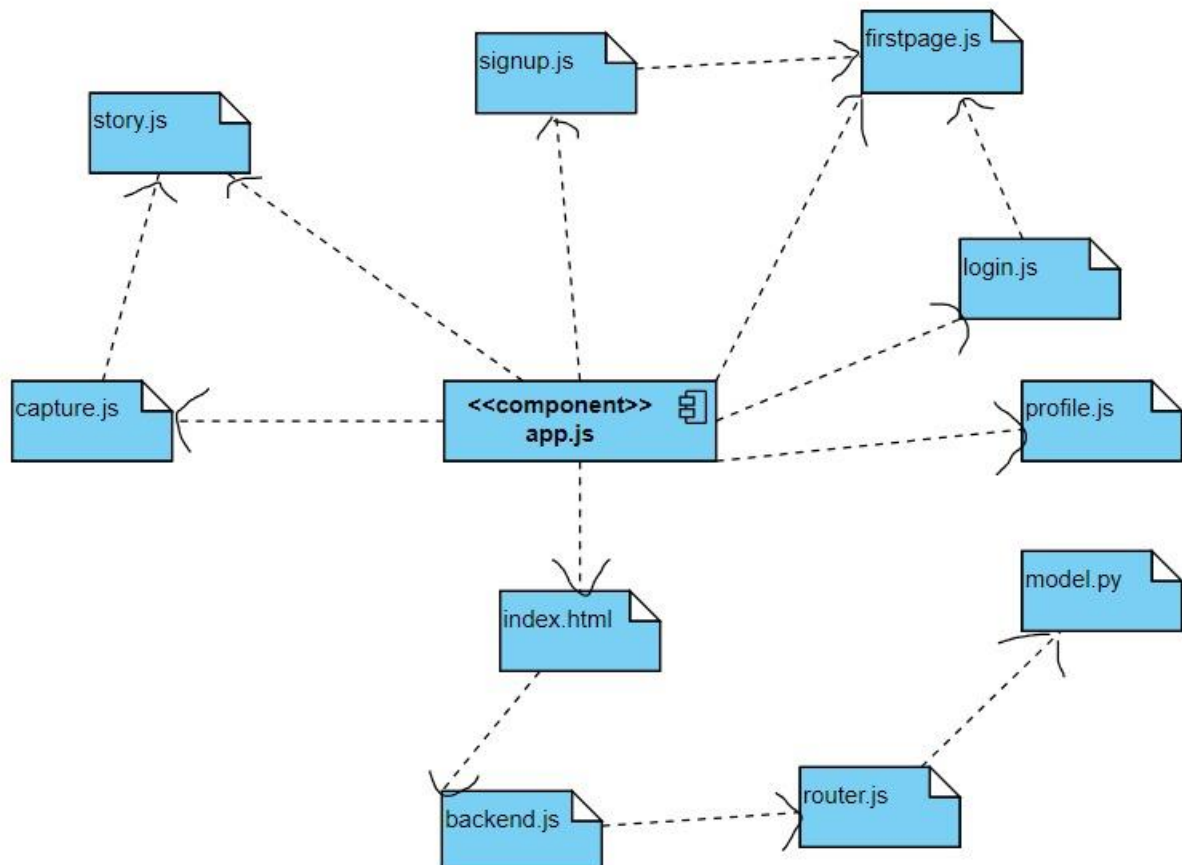
3.4 State Chart Diagram

3.4.1 Login

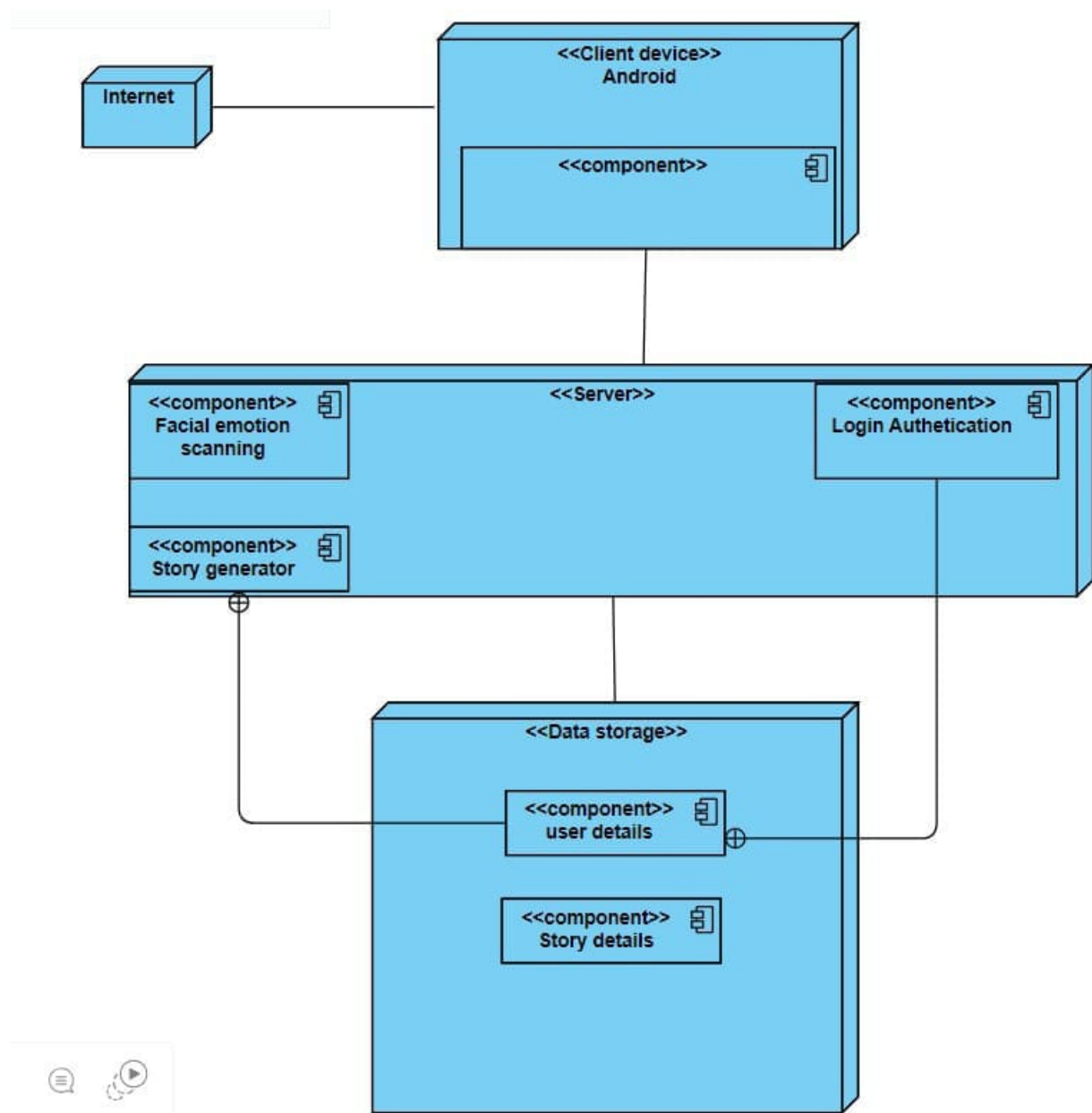


4 Implementation

4.1 Component Diagram



4.2 Deployment Diagram

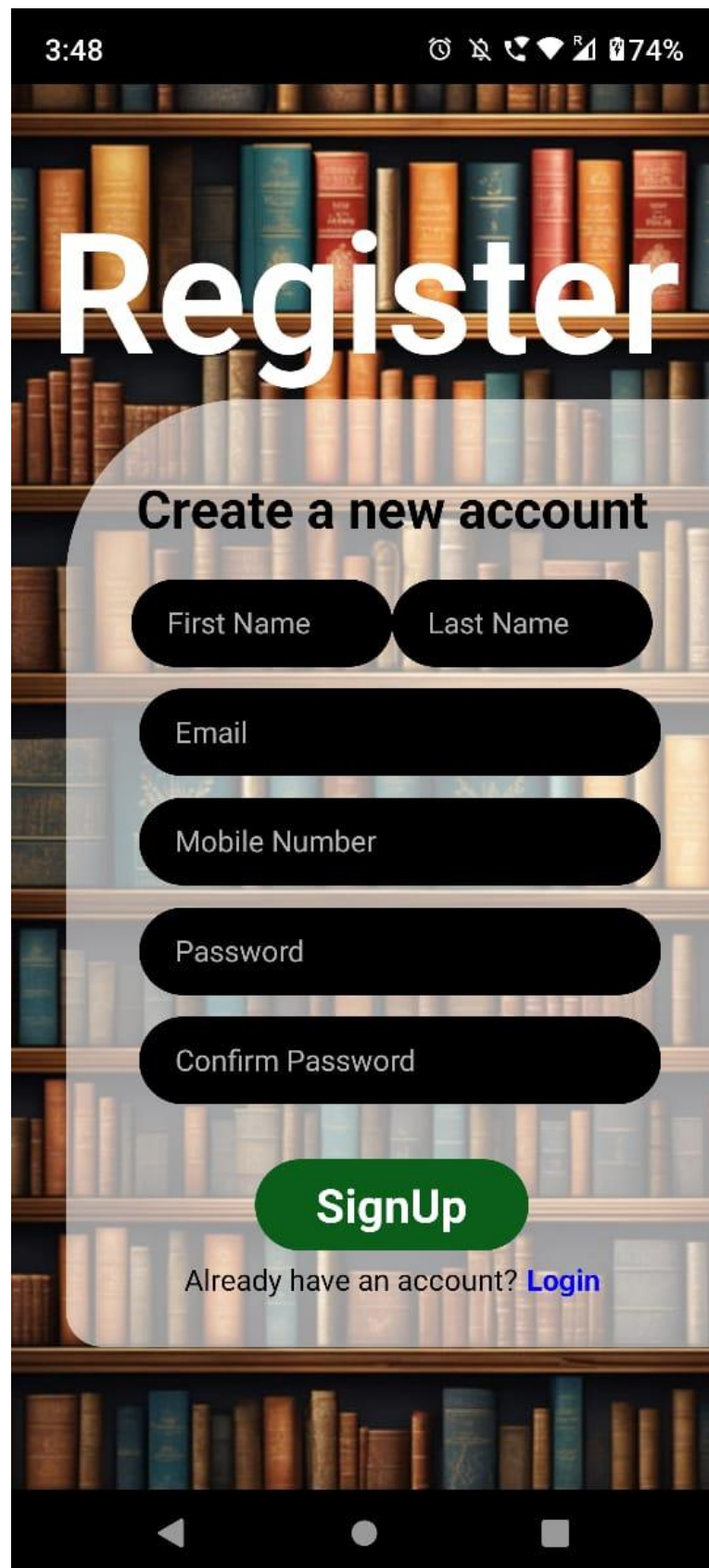


4.3 Screenshots

Landing Page



Registration Page

A mobile app registration page with a background image of a bookshelf. The page features a large 'Register' title, a 'Create a new account' subtitle, and several input fields for user information. A green 'SignUp' button is at the bottom, followed by a link to 'Login' for existing users. The top status bar shows the time as 3:48 and 74% battery. The bottom navigation bar is visible at the very bottom.

3:48 74%

Register

Create a new account

First Name Last Name

Email

Mobile Number

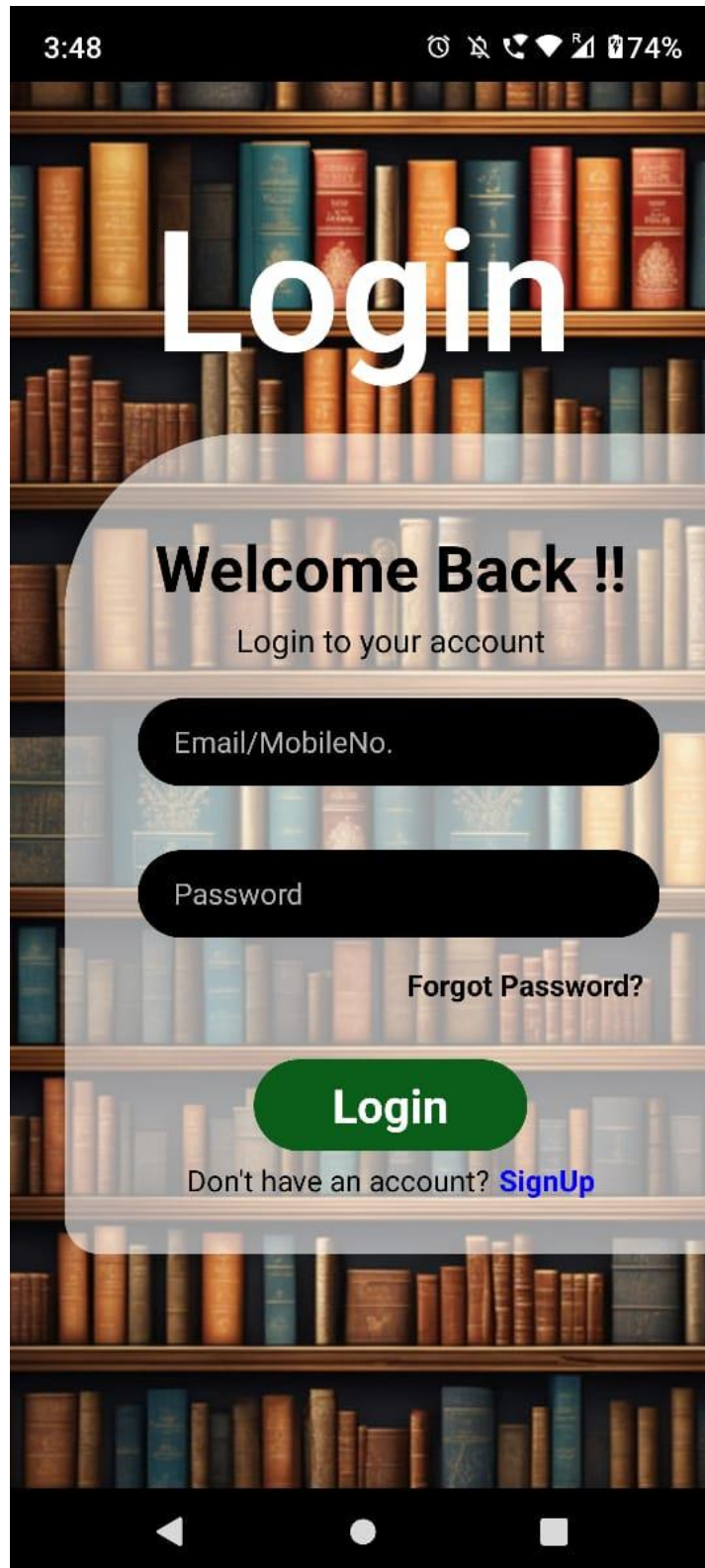
Password

Confirm Password

SignUp

Already have an account? [Login](#)

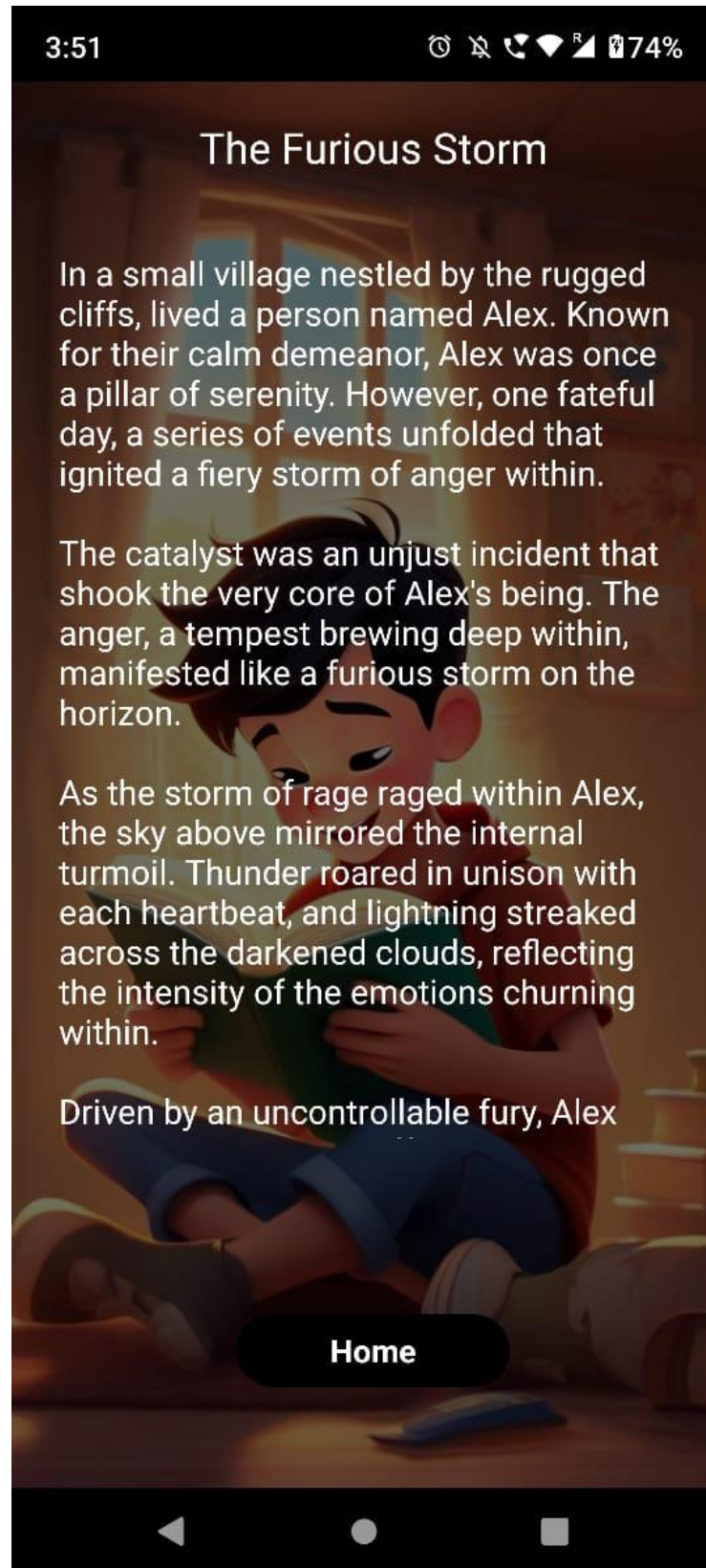
Login Page



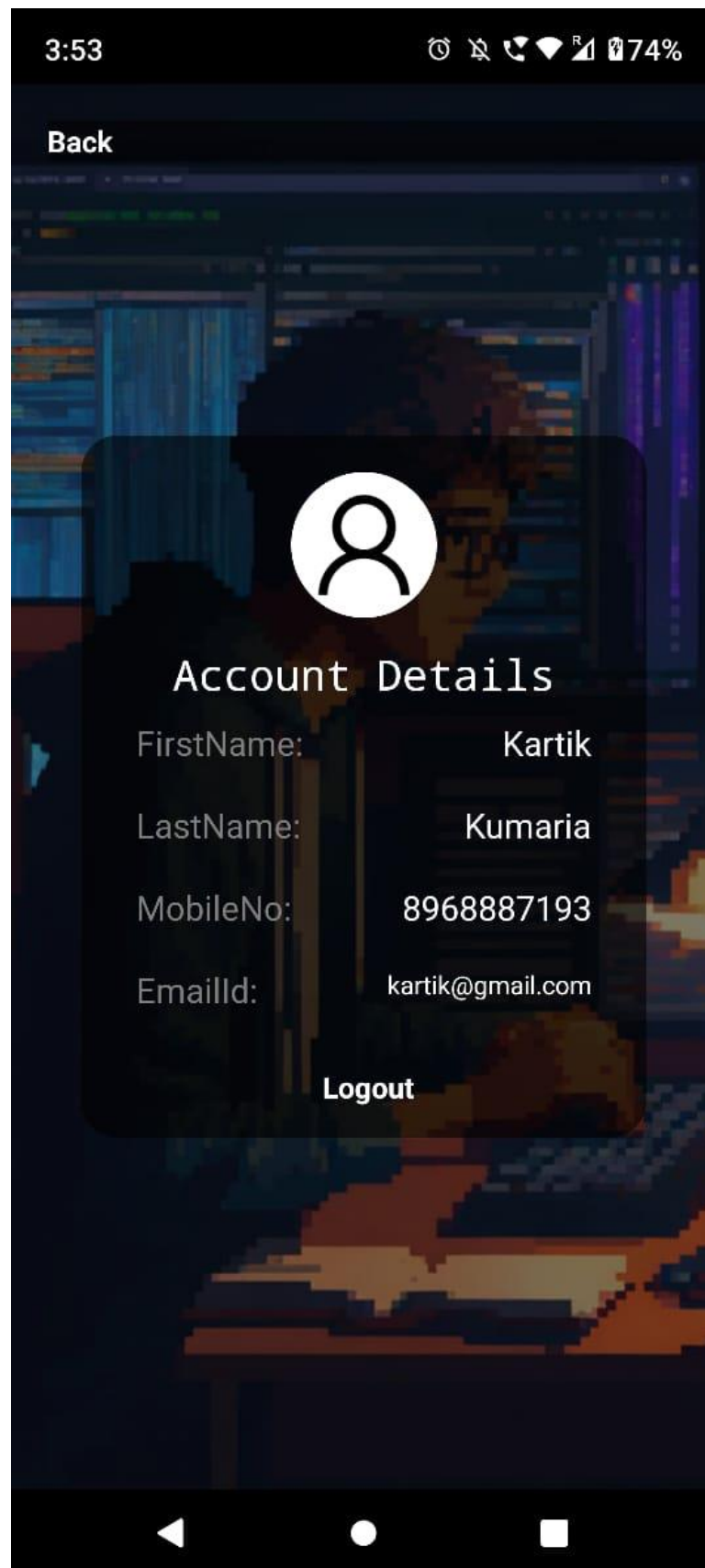
Home Page



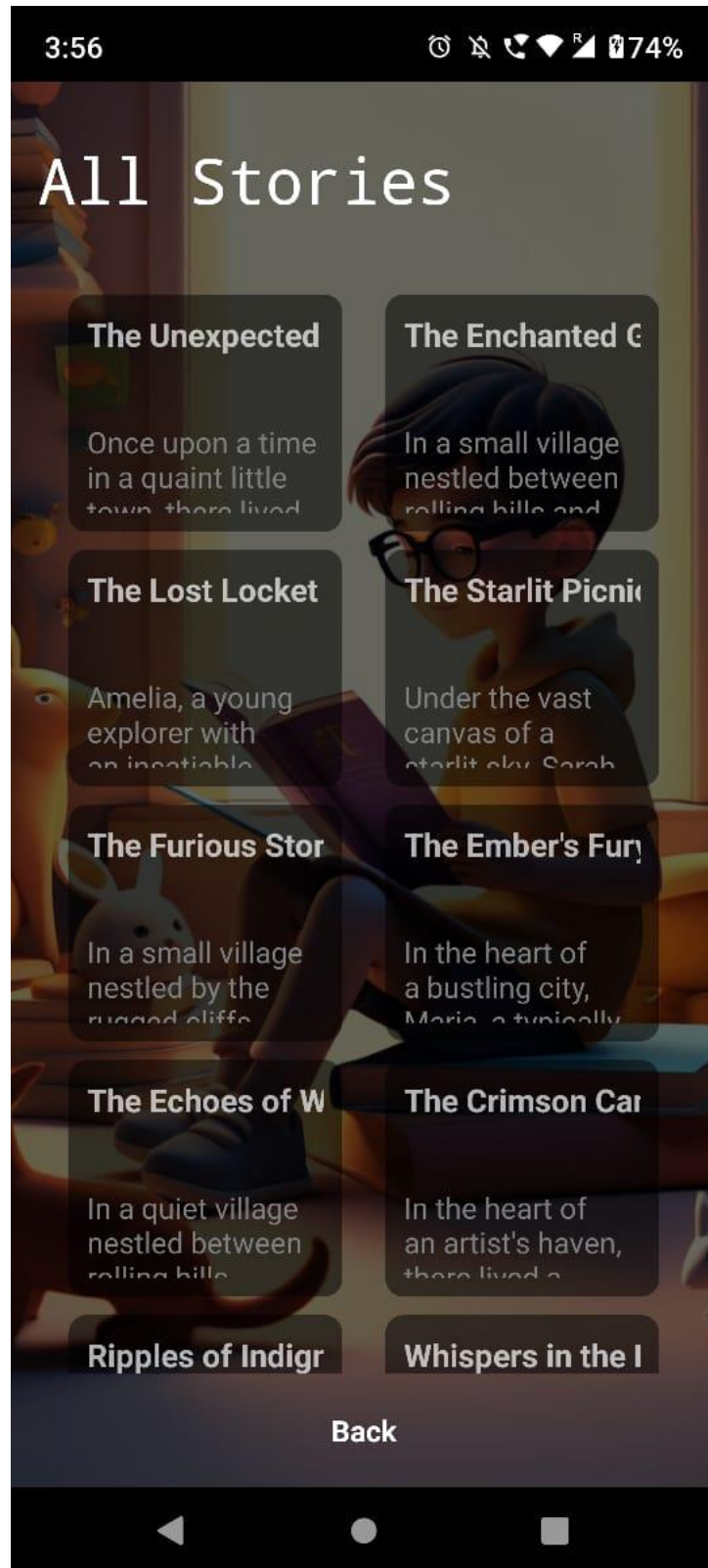
Story Display Page



User Account Details/ Profile Page



All Stories Page



5. Testing

5.1 Test Plan

Introduction:

This test plan outlines the testing strategies, test cases and methodologies that will be used to validate the Emotion Based Story Telling App.

Objectives:

The primary objective of this test plan are to:

- Ensure the application meets functional and non-functional requirements.
- Verify that the application is user friendly and easy to navigate
- Ensure that the application is secure and the user data is protected
- Ensure that the application performs well under normal and boundary conditions

Scope:

The scope of this test plan includes testing the following aspects of the Emotion Based Story Telling Application

- User Authentication
- Performance Testing
- Security testing

Testing Strategies:

The following test strategies will be used to test the DSA Tracker and Contest Calendar website:

- **Functional testing:** This will be done to ensure that all the functions of the website are working as expected.
- **Usability testing:** This will be done to ensure that the website is user-friendly and easy to navigate.
- **Compatibility testing:** This will be done to ensure that the website is compatible with different devices and browsers.
- **Performance testing:** This will be done to ensure that the website performs well under normal and peak traffic conditions.
- **Security testing:** This will be done to ensure that the website is secure and that user data is protected.

5.2 Test Cases:

The following test cases will be executed during testing:

SignUp

- 1.1 Check that user with missing signup form is unable to Signup
- 1.2 Checking that user with incorrect signup values is rejected
- 1.3 Checking that one user is not able to register twice

Login

- 2.1 Checking that user enters correct values in the required fields of the login form
- 2.2 Checking that user with incorrect values is unable to login

Emotion Detection

- 3.1 Verify that accurate Emotion Detected
- 3.2 Verify that if no face detected, required alert is produced

Story Telling

- 4.1 Verify that story is correctly displaying according to the emotion

Logout

- 5.1 Verify that user is successfully able to logout

Profile Details

- 6.1 Verify that all the user details are begin displayed correctly

All Stories

- 7.1 Verify that the user is able to read any story under the all story section

Test Methodologies:

The following test methodologies will be used during testing:

Manual testing: This will be done to test the website's functionality and usability.

The following environment will be used during testing:

Operating System: Windows 10

Devices: Samsung A22, Moto G54

Tools: Thunder Client, Nodemon , Watchman

Test Deliverables:

The following deliverables will be produced as part of this test plan:

Test cases

Test results

Test reports

5.3 Test Reports

Test Case #1.1

Test Case #: 1.1	Test Case Name: Missing Values
System: Facial Emotion recognition & StoryTeller	Subsystem: SignUp
Designed by: Siddharth Maithani	Design Date: 25/11/23
Executed by: Kartik Kumaria	
Short Description: Test the Missing Values Validation in Signup	

Pre-conditions

The User is on the Signup page

Step	Action	Expected System Response	Pass/ Fail	Comment
1	Fill in the values in the form	The values are filled in the respective field on each change		
2	Some/All input Fields are left empty	Show Alert to the user to that fields are missing		
3	Check Post-Condition 1)			

Post-conditions

1) User is not registered

Test Case #1.2

Test Case #: 1.2	Test Case Name: Incorrect Values
System: Facial Emotion recognition & StoryTeller	Subsystem: SignUp
Designed by: Siddharth Maithani	Design Date: 25/11/23
Executed by: Kartik Kumaria	
Short Description: Test the Incorrect Values Validation in Signup	

Pre-conditions

The User is on the Signup page

Step	Action	Expected System Response	Pass/ Fail	Comment
1	Fill in the values in the form	The values are filled in the respective field on each change		
2	Email is not in correct format	Show Alert to the user that email is incorrect		
3	Check Post-Condition 1)			
4	Password & confirm doesnot match	Show Alert to the User that Passwords donot match		
5	Check Post-Condition 1)			
6)	Password Entered is not in range	Show Alert to the user that password is out of range		
7)	Check Post-Condition 1)			

Post-conditions

1) User is not registered & the incorrect fields are reset.

Test Case #1.3

Test Case #: 1.3

Test Case Name: Duplicate Registration

System: Facial Emotion recognition & StoryTeller

Subsystem: SignUp

Designed by: Siddharth Maithani

Design Date: 25/11/23

Executed by: Kartik Kumaria

Short Description: Test the Duplicate Registration in Signup

Pre-conditions

The User is on the Signup page

Step	Action	Expected System Response	Pass/ Fail	Comment
1	Fill in the values in the form	The values are filled in the respective field on each change		
2	Email is already used for registration	Show Alert to the User		
3	Check Post-Condition 1)			
4	Mobile Number is already registered	Show Alert to the User		
5	Check Post-Condition 1)			

Post-conditions

1) User is not registered

Test Case #2.1

Test Case #: 2.1

Test Case Name: Incorrect Login Values

System: Facial Emotion recognition & StoryTeller

Subsystem: Login

Designed by: Siddharth Maithani

Design Date: 25/11/23

Executed by: Kartik Kumaria

Short Description: Test the Incorrect Values in Login

Pre-conditions

The User is on the Login page

Step	Action	Expected System Response	Pass/ Fail	Comment
1	Fill in the values in the form	The values are filled in the respective field on each change		
2	Password is out of range	Show alert to the user that password is out of range		
3	Check Post-Condition 1)			

Post-conditions

1) User is not Logged In

Test Case #2.2

Test Case #: 2.2

Test Case Name: Invalid Login info

System: Facial Emotion recognition & StoryTeller

Subsystem: Login

Designed by: Siddharth Maithani

Design Date: 25/11/23

Executed by: Kartik Kumaria

Short Description: Test Invalid Account info in Login

Pre-conditions

The User is on the Login page

Step	Action	Expected System Response	Pass/ Fail	Comment
1	Fill in the values in the form	The values are filled in the respective field on each change		
2	Invalid Account Details are entered	User not found is displayed to the user and values are reset		
3	Check Post-Condition 1)			

Post-conditions

1) User is not Logged In

Test Case #3.1

Test Case #: 3.1

Test Case Name: Accurate emotion detection

System: Facial Emotion recognition & StoryTeller

Subsystem: Face Emotion Detection

Designed by: Kartik Kumaria

Design Date: 25/11/23

Executed by: Siddharth Maithani

Short Description: Accurate face emotions are detected are displayed

Pre-conditions

The User is on the Login page

Step	Action	Expected System Response	Pass/ Fail	Comment
1	Face Picture is clicked	The value is sent to the model		
2	Prediction of the image is performed	Emotion predicted is displayed to the user		
3	Check Post-Condition 1)			

Post-conditions

1) Emotion is displayed to the user

Test Case #3.2

Test Case #: 3.2

Test Case Name: Faceless Pictures Detection

System: Facial Emotion recognition & StoryTeller

Subsystem: Face Emotion Detection

Designed by: Kartik Kumaria

Design Date: 25/11/23

Executed by: Siddharth Maithani

Short Description: Pictures without face are handled by the system

Pre-conditions

The User is on the Login page

Step	Action	Expected System Response	Pass/ Fail	Comment
1	Faceless Picture is clicked by the user	The picture is sent to the model		
2	Prediction of the image is performed	Model detects that there is no face present		
3	Check Post-Condition 1)			

Post-conditions

1) User is notified that there is no face detected

Test Case #4.1

Test Case #: 4.1

Test Case Name: StoryTeller

System: Facial Emotion recognition & StoryTeller

Subsystem: Story Display

Designed by: Kartik Kumaria

Design Date: 25/11/23

Executed by: Siddharth Maithani

Short Description: Stories are displayed by the Storyteller module

Pre-conditions

The User is on the StoryTeller Page

Step	Action	Expected System Response	Pass/ Fail	Comment
1	The User clicks the Show Stories Button	Story is fetched from the database according to the emotion		
2	Story is Extracted from the JSON	the Reader Component displays the Story		
3	Check Post-Condition 1)			

Post-conditions

1) Story is displayed to the user

Test Case #5.1

Test Case #: 5.1

Test Case Name: Logout

System: Facial Emotion recognition & StoryTeller

Subsystem: Logout

Designed by: Kartik Kumaria

Design Date: 25/11/23

Executed by: Siddharth Maithani

Short Description: Logout logs the user out of the system

Pre-conditions

The User is on the StoryTeller Page

Step	Action	Expected System Response	Pass/ Fail	Comment
1	The User clicks on the Logout Button	The User is logged out of the system		
2	Check Post-Condition 1)			

Post-conditions

1) User is taken to the Login Page

Test Case #6.1

Test Case #: 6.1

Test Case Name: Profile Details

System: Facial Emotion recognition & StoryTeller

Subsystem: Profile

Designed by: Kartik Kumaria

Design Date: 25/11/23

Executed by: Siddharth Maithani

Short Description: Displays the User information to the logged in user

Pre-conditions

The User is logged in
The User is at the Entry Page

Step	Action	Expected System Response	Pass/ Fail	Comment
1	The Clicks on the Drop Down Menu	The Logout and Profile Options are displayed		
2	User selects the profile option	All the Details of the User in databse are displayed		
3	Check Post-Condition 1)			

Post-conditions

1) User can view their details at the profile page

Test Case #7.1

Test Case #: 7.1

Test Case Name: Select from all stories

System: Facial Emotion recognition & StoryTeller

Subsystem: Story Selector

Designed by: Siddharth Maithani

Design Date: 25/11/23

Executed by: Kartik Kumaria

Short Description: Gives user the option to select from all stories

Pre-conditions

The User is logged in
The User is at the Entry Page

Step	Action	Expected System Response	Pass/ Fail	Comment
1	The User selects the Explore Stories	Stories present in all collections are displayed		
2	User Selects the Story	The Selected story is sent to the Story Teller		
3	Check Post-Condition 1)			

Post-conditions

- 1) The User is displayed the story by storyteller